

MATH 112 – Calculus II

Fall 2025 – Programming Project Guidelines

Submission Guidelines

General Instructions

- Each question must be solved using both **Python code** and **hand-written computation**, unless explicitly marked otherwise.
- Your solutions should demonstrate a clear understanding of both analytical and computational approaches.
- All students are equally responsible for the final product. Splitting the work helps with efficiency but does not reduce accountability. Each member should review the full project and ensure it meets the required standards.

Submission Format

- Use the provided **Answer Template** to structure your submission.
- Each submission must include the following:
 - Section number and group number
 - Full names and student ID of group members with clear breakdown of each member's contribution
 - The question statement
 - Python code implementation
 - Explanation of the code logic
 - Hand-written solution (scanned and embedded)
- File naming convention:
 - PDF document: `S[SectionNumber]_G[GroupNumber].pdf`
 - Jupyter Notebook: `S[SectionNumber]_G[GroupNumber].ipynb`
 - Example:
 - * `S12_G03.pdf`
 - * `S12_G03.ipynb`
- If submitting multiple files, compress them into a single archive. Example: `S12_G03.zip`
- Ensure that the division of work is clearly documented in the report.
- All hand-written solutions must be scanned clearly:
 - Use scanning apps for better quality.
 - Avoid submitting images in `.heic` format (Apple users).
 - Ensure all images are legible and properly cropped.
- Test all files before submission to confirm they are complete and executable.
- Organize your document and code with clear headings for each question.

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Problem 1: Integration

Consider the function $f(x) = \frac{2e^x}{e^{3x} + e^x}$.

- a) [3 points, code only] Plot the graph of f .
- b) [7 points] Find the anti-derivative of f .
- c) [5 points] Find the area between the graph of $f(x)$ and the x -axis for $-2 \leq x \leq 2$.
- d) [10 points] Consider the integral $\int_{-2}^{\infty} f(x) dx$. Determine whether this integral is improper or not. Explain your answer. Determine whether the integral converges or diverges. If it is convergent, determine its value.

Problem 2: Parametric Curve, Speed, and Distance Traveled

Let

$$x(t) = 2 \cos t + \cos(2t), \quad y(t) = 2 \sin t - \sin(2t), \quad t \in [0, 2\pi].$$

- a) [5 points, code only] Plot the curve and mark the points corresponding to $t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$.
- b) [8 points] Compute $\frac{dy}{dx}$ (in terms of t). Find all $t \in [0, 2\pi]$ where the curve has horizontal tangents and where it has vertical tangents. Give the corresponding (x, y) points.
- c) [6 points] Compute the speed $v(t) = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$. Show by hand that $v(t) \geq 0$ and identify where (if anywhere) $v(t) = 0$. Interpret what $v(t) = 0$ means for the motion/curve.
- d) [6 points] Find the total distance traveled over $t \in [0, 2\pi]$ (arc length). Set up the exact integral by hand, then evaluate numerically in Python and report a decimal approximation (to 4 decimals). Briefly comment on why an elementary closed form may or may not be expected.

Problem 3: Infinite Series

- a) [12 points, code only] Plot the first 100 terms of the sequence $a_n = (-1)^n \frac{e^{-n}}{n}$, $n = 1, 2, \dots$. Does the sequence appear to converge or diverge?
- b) [13 points] Implement the **Ratio Test** on the series $\sum_{k=3}^{\infty} (-1)^k \frac{3^k}{k!}$ and determine if it converges.